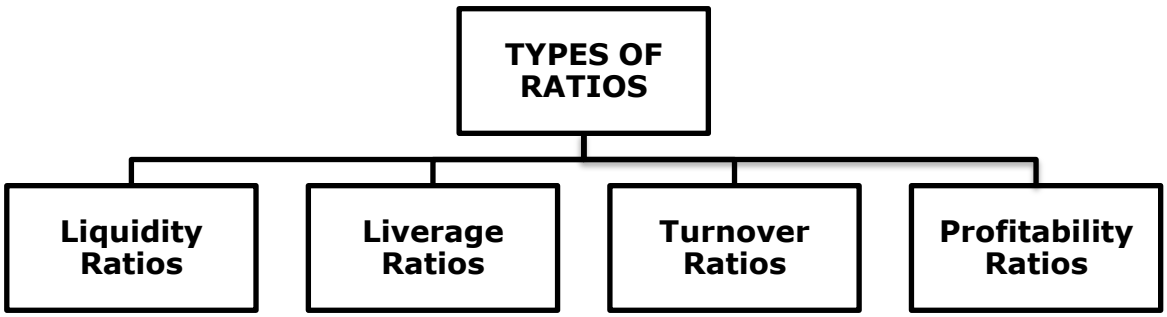
RATIOS

Absolute figures are valuable but they standing alone convey no meaning unless compared with another. Accounting ratio show inter-relationships which exist among various accounting data. When relationships among various accounting data supplied by financial statements are worked out, they are known as accounting ratios.

What is a ratio?

Ratio analysis is a means for financial analysis. Ratio is a mathematical relationship between two accounting figures. They show the relationship between two items in a more meaningful way which help us to draw certain conclusions. Ratios may be used to compare the previous data, to compare one firm with another firm etc. the ratios can be expressed as percentage or proportion or times based on the nature of ratio.



LIQUIDITY RATIOS

Liquidity ratios express the ability of the firm to meet its short-term Obligations as when they become due. Creditors are interested to know whether the firms is in a position to meet its commitments on time or not. These ratios help in identifying the danger signals for the firm in advance. The important liquidity ratios are given below.

1. Current Ratio:- It is also called as working capital ratio. It is the ratio between current assets and current liabilities. The firm is in comfortable position if its current ratio is 2:1. It means for every rupee of current liability, there should be two rupees worth of current assets.

$$\text{Current Ratio} = \frac{\text{Current Asstes}}{\text{Current Liabilties}}$$

Current assets = Cash + cash in bank + marketable securities + short term investments + bills receivables +debtors + inventory + stock + work-in-progress + pre-paid expenses + incomes receivable (accrued income) etc.

Current liabilities = Expenses payable + bills payable + creditors + short term loans + income tax to be paid + dividend payable + bank overdraft + long term loans and debentures to be paid within one year + provision for tax + short term advances etc.

2. Quick Ratio:- It is also called as working Acid test ratio or liquid ratio. It is the ratio between quick assets and current liabilities. The firm is in comfortable position if its current ratio is 1:1. It means for every rupee of current liability, there should be one rupee worth of quick assets. Quick assets can be converted into cash quickly.

$$\textbf{Quick Ratio} = \frac{\textbf{Quick Asstes}}{\textbf{Current Liabilties}}$$

Quick assets = All current assets except stock and prepaid expenses.

Example 1:

From the Balance Sheet of XYZ Co. Ltd., calculate liquidity ratios.

(Rs. in thousands)

Capital & Liabilities	Amount	Assets	Amount
Preference share capital	100	Land and Buildings	225
Equity share capital	150	Plant and machinery	250
General reserve	250	Furniture and Fixtures	100
Debentures	400	Stock	250
Creditors	200	Debtors	125
Bills payable	50	Cash at Bank	250
Outstanding expenses	50	Cash in hand	125
Profit and loss account	100	Prepaid expenses	50
Bank loan(Long term)	200	Marketable securities	125
	1500		1500

Solution:

$$\textbf{Current Ratio} = \frac{\textbf{Current Asstes}}{\textbf{Current Liabilties}}$$

Current assets = Stock + Debtors + Cash at Bank + Cash in hand + Prepaid expenses + Marketable securities.

= 250 + 125 + 250 + 125 + 50 + 125 = 925

Current liabilities = Creditors + Bills payable + Outstanding expenses.

= 200+ 50 + 50 = 300

$$\textbf{Current Ratio} = \frac{925}{300} = 3.08:1$$

Quick assets = Debtors + Cash at Bank + Cash in hand + Marketable securities.

= 125 + 250 + 125 + 125 = 625

$$\textbf{Quick Ratio} = \frac{625}{300} = 2.08:1$$

ACTIVITY RATIOS/TURNOVER RATIOS

Activity ratios are classed as Turnover ratios. These ratios tell how active the firm is in selling stocks, collecting money from debtors and paying to creditors. They are given below.

1. Inventory Turnover Ratio:- It is also called as Stock turnover ratio. It indicates the number of times the average stock is being sold during a given accounting period. The higher the ratio, the better is the performance of the firm in selling its stock. It is the rate at which inventories are converted into sales and then to cash.

$$\textbf{a) Inventory Turnover Ratio} = \frac{\textbf{Cost of goods sold}}{\textbf{Average Stock}}$$

Cost of goods sold = Opening stock + Purchases + Manufacturing expenses – Closing stock

(or)

= Sales – Gross profit

$$\textbf{Average stock} = \frac{\text{Opening stock} + \text{Closing stock}}{2}$$

Note 1. When cost of goods sold is not given, sales amount should be taken into account.

2. When opening stock is not given, closing stock is considered as 'average stock'.

$$\textbf{b) Inventory Holding Period} = \frac{\textbf{365 days}}{\textbf{Inventory Turnover Ratio}}$$

Example 1:

A firm sold goods worth Rs. 500000 and its gross profit is 20% of sales value. The inventory at the beginning of the year was Rs. 16000 and at end of the year was 14000. Compute inventory turnover ratio and also the inventory holding period.

Solution:

$$\textbf{a) Inventory Turnover Ratio} = \frac{\text{Cost of goods sold}}{\text{Average Stock}} = \frac{400000}{15000} = 26.66 \text{ times}$$

Cost of goods sold = Sales – Gross profit = 500000 – (500000 x 20%) = 400000

$$\textbf{Average stock} = \frac{\text{Opening stock} + \text{Closing stock}}{2} = \frac{16000 + 14000}{2} = 15000$$

$$\textbf{b) Inventory Holding Period} = \frac{365 \text{ days}}{\text{Inventory Turnover Ratio}} = \frac{365}{26.66} = 14 \text{ days}$$

2. Debtors Turnover Ratio:- It reveals the number of times the average debtors are collected during a given accounting period. The firms usually prepare the aged list of debtors showing the details of when to collect and how much to collect from debtors. The higher the ratio, the better is the performance of the firm in collecting money from debtors.

$$\textbf{a) Debtors Turnover Ratio} = \frac{\textbf{Net credit sales}}{\textbf{Average Debtors}}$$

Net Credit Sales = Credit sales – Returns

Note: 1. When credit sales are not given, total sales are taken.

$$\textbf{Average debtors} = \frac{\text{Opening debtors} + \text{Closing debtors}}{2}$$

Note: 1. If opening debtors are not given, closing debtors should be considered as average debtors.

$$\textbf{b) Debt Collection Period} = \frac{\textbf{365 days}}{\textbf{Debtors Turnover Ratio}}$$

Example:

A firm's sales during the year was Rs. 400000 of which 60% were credit sales. The balance of debtors at the beginning and ending year were 25000 and 15000 respectively. Calculate debtors turnover ratio of the firm. Also find out debt collection period.

Solution:

$$\textbf{a) Debtors Turnover Ratio} = \frac{\text{Net credit sales}}{\text{Average Debtors}} = \frac{240000}{20000} = 12 \text{ times}$$

Net credit sales = Sales x 60/100 = 400000 x 60/100 = 240000

$$\textbf{Average debtos} = \frac{\text{Opening debtors} + \text{Closing debtors}}{2} = \frac{25000 + 15000}{2} = 20000$$

$$\textbf{b) Debt Collection Period} = \frac{365 \text{ days}}{\text{Debtors Turnover Ratio}} = \frac{365 \text{ days}}{12} = 30.41 \text{ days}$$

3. Creditors Turnover Ratio:- It reveals the number of times the average creditors are paid during a given accounting period. The firms usually prepare the aged list of creditors showing the details of when to pay and how much to pay to its creditors. It shows how promptly the firm is in a position to pay its creditors.

$$\textbf{a) Creditors Turnover Ratio} = \frac{\textbf{Net credit purchases}}{\textbf{Average Creditors}}$$

Net Credit Purchases = Credit Purchases – Returns

Note: 1. When credit purchases are not given, total purchases are taken.

$$\textbf{Average creditors} = \frac{\text{Opening creditors} + \text{Closing creditors}}{2}$$

Note: 1. If opening creditors are not given, closing creditors should be considered as average creditors.

$$\textbf{b) Creditors Payment Period} = \frac{\textbf{365 days}}{\textbf{Creditors Turnover Ratio}}$$

Example:

A firm's purchases during the year was Rs. 400000 of which 50% were credit purchases. The balance of creditors at the beginning and ending year were 30000 and 10000 respectively. Calculate creditors turnover ratio of the firm. Also find out creditors payment period.

Solution:

$$\textbf{a) Creditors Turnover Ratio} = \frac{\text{Net credit purchases}}{\text{Average Creditors}} = \frac{200000}{20000} = 10 \text{ times}$$

Net credit purchases = Purchases x 50/100 = 400000 x 50/100 = 200000

$$\textbf{Average creditors} = \frac{\text{Opening creditors} + \text{Closing creditors}}{2} = \frac{30000 + 10000}{2} = 20000$$

$$\textbf{b) Creditors Payment Period} = \frac{365 \text{ days}}{\text{Creditors Turnover Ratio}} = \frac{365 \text{ days}}{10} = 36.5 \text{ days}$$

CAPITAL STRUCTURE RATIOS:

Capital structure ratios are also called as leverage ratios. These ratios focus on the long term solvency of the firm. The long term solvency of the firm always reflected in its ability to meet its long term commitments such as payment of interest periodically without fail, repayment of principal as and when the become due. The below are the most commonly used capital structure rations.

1. **Debt-Equity Ratio**:- It is the ratio between outsider's funds(Debt) and insider's funds (Equity). It is a measure of solvency. This ratio is used to measure the firm's obligations to creditors in relation to the owners' funds. The standard ratio is 1:1. this means for every rupee of debt, there should be one rupee worth internal funds. A high D/E ratio implies that the creditors stake is more as compared to that of owners.

$$\textbf{Debt} - \textbf{Equity Ratio} = \frac{\textbf{Debt}}{\textbf{Equity}} = \frac{\textbf{Outsiders Funds}}{\textbf{Insider's Funds}} = \frac{\textbf{Long term loans}}{\textbf{Shareholders Funds or Net worth}}$$

Debt = Debentures + bonds + mortgage loan + other long term loans.

Equity = Equity share capital + preference share capital + capital reserve + revenue reserve + sinking fund + contingent reserve – artificial assets.

Note: Artificial assets = preliminary expenses + deferred revenue expenses + discount on issue of shares/ debentures + profit and loss A/C debit balance + underwriting commission

. **Example 1:** Calculate Debt – Equity ratio from the following data.

Debentures Rs. 400000, Long term loans Rs. 200000, Preference share capital Rs. 100000, Equity share capital Rs. 150000, General reserve Rs. 250000, Profit & Loss account Rs. 100000.

Solution:

$$\textbf{Debt} - \textbf{Equity Ratio} = \frac{\text{Debt}}{\text{Equity}} = \frac{600000}{600000} = 1:1$$

Debt = 400000 + 200000 = 600000

Equity = 100000 + 150000 + 250000 + 100000 = 600000

2. Interest Coverage Ratio:- This ratio judges the firm's capacity to pay the interest on debt it borrows. The higher the ratio, better it is. A ratio implies that the company has no problems in paying interest.

$$\textbf{Interest Coverage Ratio} = \frac{\textbf{EBIT}}{\textbf{Interest}} = \frac{\textbf{PBIT}}{\textbf{Interest}}$$

EBIT = Earnings Before Interest and Tax

PBIT = Profit Before Interest and Tax

Interest = Fixed interest on long term loans

Example: EBIT of a company is Rs. 560000. Its fixed commitments include payment of 10 percent on 7000 debentures of Rs. 100 each. It is subject to tax of 30 percent per annum. Calculate interest coverage ratio.

Solution:

$$\textbf{Interest Coverage Ratio} = \frac{\text{EBIT}}{\text{Interest}} = \frac{560000}{70000} = 8 \text{ times}$$

EBIT = Rs. 560000

Debentures Amount = 7000 debentures x Rs. 100 each = Rs. 700000

Fixed interest charges on debentures = 700000 x 10/100 = 70000

3. Proprietor's Funds to Total Assets Ratio

Proprietors' Funds = Equity share capital + Preference share capital + General reserve + Employee Provident Fund + profit and loss account.

Total Assets = Tangible assets and Current assets

$$\textbf{Propietors}'\textbf{Funds to Total Assets Ratio} = \frac{\textbf{Proprietors}'\textbf{Funds}}{\textbf{Total Assets}}$$

Proprietors' Funds = Equity share capital + Preference share capital + General reserve + Employee Provident Fund + profit and loss account.

Total Assets = Tangible assets and Current assets

Example 1:

From the Balance Sheet of XYZ Co. Ltd., calculate liquidity ratios.

(Rs. in thousands)

Capital & Liabilities	Amount	Assets	Amount
Preference share capital	100	Land and Buildings	225
Equity share capital	150	Plant and machinery	250

General reserve	250	Furniture and Fixtures	100
Debentures	400	Stock	250
Creditors	200	Debtors	125
Bills payable	50	Cash at Bank	250
Outstanding expenses	50	Cash in hand	125
Profit and loss account	100	Prepaid expenses	50
Bank loan(Long term)	200	Marketable securities	125
	1500		1500

Solution:

$$\textbf{Propietors'Funds to Total Assets Ratio} = \frac{\text{Proprietors' Funds}}{\text{Total Assets}} = \frac{600}{1500} \text{x } 100 = 40\%$$

Proprietors' Funds to Total Assets = 100+150+250+100=600

Total assets = 225+250+100+250+125+250+125+50+125=1500

PROFITABILITY RATIOS

Profitability ratios indicate how well the firm is operating its Activities in a profitability manner. Owners want a reasonable rate of return on their investment. So, the firm has to generate profits to meet the expectations of shareholders and also for further expansion of the business. The following are the common profitability ratios.

1. **Gross Profit Ratio**:- It is the ratio between gross profit and net sales. It is expressed in percentage.

$$\textbf{Gross Profit Ratio} = \frac{\textbf{Gross Profit}}{\textbf{Net sales}} \textbf{x 100}$$

Gross profit = Net sales – Cost of goods sold

(or)

=(sales+closing stock)-(opening stock+purhases)

Net sales = Total sales – Sales returns

Cost of goods sold = Opening stock + Net purchases + Production expenses +Closing sotck

(or)

= Net sales – Gross profit

Example: Net sales is Rs. 50000 for a firm and cost of goods sold is Rs. 20000. Calculate gross profit ratio.

Solution:

$$\textbf{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Net sales}} \text{x } 100 = \frac{30000}{50000} \text{x } 100 = 60\%$$

Gross profit = Net sales – Cost of goods sold = 50000-20000 = 30000

2. Net Profit Ratio:- It is the ratio between net profit after tax and net sales. It is expressed in percentage.

$$\textbf{Net Profit Ratio} = \frac{\textbf{Net Profit after tax}}{\textbf{Net sales}} \textbf{x 100}$$

Net Profit after Tax = (Operating Profit + Non-operating Income) – (Non-operating Expenses + Taxes)

Operating Profit = (Net Sales – Operating Cost)

Operating Cost = (Cost of goods sold + Operating expenses)

Operating Expenses = (Office and Administration expenses + Sales and Distribution expenses)

(or)

Net Profit after Tax = Gross profit – All expenses and losses + All incomes –Tax

Example : Calculate net profit ratio from the following data.

Net sales Rs. 50000
Cost of goods sold Rs. 20000
Administration Expenses Rs. 3000
Selling and Distribution expenses Rs 4000
Loss on sale of fixed assets Rs. 3000
Interest on investment received Rs. 2000
Tax 20%

Solution:

Particulars	Rs	Rs
Sales	50000	
Less: Cost of goods sold	20000	
Gross Profit		30000
Less: Administration expenses	3000	
Selling and Distribution expenses	4000	7000
Net Profit		23000
Add: Interest on investments		2000
		25000
Less: Loss on sale of Asset		3000
		22000
Tax 20% (22000x20/100)		4400
Net Profit After Tax		17600

$$\textbf{Net Profit Ratio} = \frac{\text{Net Profit after tax}}{\text{Net sales}} \text{x } 100 = \frac{17600}{50000} \text{x} 100 = 35.2\%$$

3. Operating Ratio:- It is the ratio between cost of goods sold plus operating expenses and net sales. It is expressed as percentage to sales.

$$\textbf{Operating Ratio} = \frac{\textbf{Operating cost}}{\textbf{Net sales}} \textbf{ x 100}$$

Operating Cost = (Cost of goods sold + Operating expenses)

Operating Expenses = (Office and Administration expenses + Sales and Distribution expenses)

Operating Profit Ratio = 100 – Operating Ratio

Example: Calculate operating ratio from the following data.

Net sales Rs. 50000
Cost of goods sold Rs. 20000
Administration Expenses Rs. 3000
Selling and Distribution expenses Rs 4000
Loss on sale of fixed assets Rs. 3000
Interest on investment received Rs. 2000
Tax 20%

Solution:

$$\textbf{Operating Ratio} = \frac{\text{Operating cost}}{\text{Net sales}} \text{x } 100 = \frac{27000}{50000} \text{x} 100 = 54\%$$

Operating cost = (cost of goods sold + Operating Expenses)
= (20000+3000+4000) = 27000

4. Return On Investment (ROI):- This ratio is also called as Return On Capital Employed (ROCE). The firm is interested to assess the return on capital employed.

$$\textbf{Return On Investement} = \frac{\textbf{PBIT}}{\textbf{Net Assets (or) Capital Employed}} \textbf{x } 100$$

Profit Before Interest and Tax (PBIT) = Gross profit – All expenses and losses + All incomes

Capital employed = Equity share capital + Preference share capital + Reserves + Long term loans + Debentures – Intangible assets

(or)

= Fixed assets + Current assets – Current liabilities

5. Return On Equity (ROE):- The equity shareholders are interested to assess the return on equity capital employed.

$$\textbf{Return On Equity} = \frac{\textbf{PAT} - \textbf{Preference dividend}}{\textbf{Equity shareholders' Funds}} \textbf{x } 100$$

Equity Shareholders Funds = Equity Share capital + Reserves and Surpluses

6. Earnings Per Share (EPS):- EPS is the relationship between net profit and the number of equity shares outstanding at eth end of the given period.

$$\textbf{Earnings Per Share} = \frac{\textbf{PAT} - \textbf{Preference dividend}}{\textbf{No. of equity shares}}$$

Example 1: Given that the number of share is 10000 and the net profit after taxes for a given period is Rs. 450000, the EPS can be calculated as follows:

Solution:

$$\textbf{Earnings Per Share} = \frac{\textbf{PAT} - \textbf{Preference dividend}}{\textbf{No. of equity shares}} = \frac{\textbf{450000} - \textbf{0}}{\textbf{10000}} = \textbf{Rs. 45}$$

7. Dividend Yield Ratio (D/Y Ratio):- Yield means the amount of total return the investor will receive for a given period of time for the amount of his investment. Dividend yield refers to the percentage return on he price paid for shares. It is calculated as given below:

$$\text{Dividend Yield Ratio} = \frac{\text{Dividend Per Share}}{\text{Market Value Per Share}} \text{ x } 100$$

Example: Given that current market price of a share Rs. 300; face value of the share is Rs. 100; percentage of dividend declared is 20%, the yield is;

Solution:

$$\text{Dividend Yield Ratio} = \frac{\text{Dividend Per Share}}{\text{Market Value Per Share}} \text{ x } 100 = \frac{20}{300} \text{ x } 100 = 6.67\%$$

Dividend Per Share = Face value of share * 20/100 = 100 * 20/100 = 20

8. Price Earnings Ratio (P/E Ratio):- This is the ratio of the market value of a share and Earnings Per Share.

$$\text{Price} - \text{Earnings Ratio} = \frac{\text{Market Value of Equity Share}}{\text{Earnings per Share}}$$

Example: Given that market price of a share is Rs. 340 and EPS is 10, calculate P/E ratio.

Solution:

$$\text{Price} - \text{Earnings Ratio} = \frac{\text{Market Value of Equity Share}}{\text{Earnings per Share}} = \frac{340}{10} = 34$$

Problem 1:- The following an extract of a balance sheet of a company during the last year. Compute current ratio and quick ratio.

Land and buildings	50000	Plant and machinery	100000
Furniture and fixtures	25000	Closing stock	25000
Sundry debtors	12500	Wages prepaid	2500
Sundry creditors	8000	Rent outstanding	2000

Solution:

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Current assets} = \text{Sundry debtors} + \text{Closing stock} + \text{Wages prepaid.}$$

$$= 12500+25000+2500 = 40000$$

$$\text{Current liabilities} = \text{Sundry creditors} + \text{Rent outstanding}$$

$$= 8000+2000 = 1000$$

$$\text{Current Ratio} = \frac{40000}{10000} = 4:1$$

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

Quick assets = Sundry debtors
= 12500

Current liabilities = Sundry creditors + Rent outstanding
= 8000+2000= 10000

$$\text{Quick Ratio} = \frac{12500}{10000}$$

Quick Ratio = 1.25:1

Problem 2:- Calculate inventory turnover ratio and Average period of holding the stocks.

Sundry debtors	45000	Closing stock	30000
Sales	400000	Sales returns	20000
Stock as on 1-1-2014	40000	Stock as on 31-12-2014	60000

Solution:

a) $$\text{Inventory Turnover Ratio} = \frac{\text{Cost of goods sold}}{\text{Average stock}}$$

Cost of goods sold = Opening stock + purchases + manufacturing expenses- closing stock

(or)

= Sales – Gross Profit

$$\text{Average stock} = \frac{\text{Opening stock} + \text{Closing stock}}{2}$$

Note:- 1. When cost of goods sold is not given, sales amount should be taken into account.

a) **Inventory Turnover Ratio** = $\frac{\text{Cost of goods sold}}{\text{Average stock}}$

$$\text{Average stock} = \frac{\text{Opening stock} + \text{Closing stock}}{2} = \frac{40000 + 60000}{2} = \text{Rs. } 50000$$

$$\text{Inventory Turnover Ratio} = \frac{400000}{50000} = 8 \text{ times}$$

b) **Inventory holding period** = $\frac{365 \text{ days}}{\text{Inventory Turnover Ratio}} = \frac{365}{8} = 46 \text{ days}$

Problem 3:- Given the following data, calculate debtors and creditors turnover ratios.

Debtors as on 1-1-2014	8000	Debtors as on 31-12-2014	16000
Creditors as on 1-1-2014	32000	Purchases (60% credit)	150000
Furniture and fixtures	25000	Cash	5000
Creditors as on 31-12-2015	26000		
Sales (75% credit)	250000		

Solution:

a) **Debtors Turnover Ratio** = $\frac{\text{Net credit sales}}{\text{Average debtors}}$

Net credit sales = Sales x 60/100 = 250000 x 75/100 = 187500

Average debtors = (Opening debtors + closing debtors) /2
= (8000 + 16000) /2 =12000

$$\text{Debtors Turnover Ratio} = \frac{187500}{12000} = 15.6 \text{ times}$$

b) **Debtors Collection Period** = $\frac{365 \text{ days}}{\text{Debtors Turnover Ratio}} = \frac{365}{15.6} = 23.36 \text{ days}$

Contd

a) $$\text{Creditors Turnover Ratio} = \frac{\text{Net credit purchases}}{\text{Average Creditors}}$$

$$\text{Net credit purchases} = \text{Purchases} \times 60/100 = 150000 \times 60/100 = 90000$$

$$\text{Average creditors} = (\text{Opening creditors} + \text{closing creditors})/2$$
$$= (32000 + 26000)/2 = 29000$$

$$\text{Creditors Turnover Ratio} = \frac{90000}{29000} = 3.1 \text{ times}$$

b) $$\text{Creditors payment Period} = \frac{365 \text{ days}}{\text{Creditors Turnover Ratio}} \quad \frac{365}{3.1} = 117.74 \text{ days}$$

Problem 4:- Given the following data, calculate current ratio and quick ratio

Capital	360000	Debentures	420000
Reserve fund	240000	Creditors	36000
Bank over draft	60000	Rent outstanding	6000
Provision for taxation	78000	Land and buildings	440000
Plant and machinery	235000	Furniture and fixtures	140000
Motor vehicles	105000	Stock	60000
Sundry debtors	90000	Short term investments	75000
Cash at bank	30000	Cash in hand	25000

Solution:

Current assets = Sundry debtors + Cash at bank + Stock + Short term investments + cash in hand

= 90000+30000+60000 +75000 + 25000 = 280000

Current liabilities = Creditors + Rent outstanding + Bank over draft + Provision for taxation

= 36000+6000 + 60000 + 78000 = 180000

$$\text{Current Ratio} = \frac{\text{Current assets}}{\text{Current Liabilities}} = \frac{280000}{180000} = 1.55:1$$

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

Quick assets = Sundry debtors + Cash at bank + Short term investments + cash in hand

= 90000+30000+75000 + 25000 = 220000

Current liabilities = Creditors + Rent outstanding + Bank over draft + Provision for taxation

= 36000+6000 + 60000 + 78000 = 180000

$$\text{Quick Ratio} = \frac{220000}{180000}$$

$$\text{Quick Ratio} = 1.22:1$$

Problem 5:- Given the following data, calculate Debt-equity ratio, Interest coverage ratio and Proprietary funds to total assets ratio.

Liabilities and Capital	Rs	Assets	Rs
Share Capital:		Motor vehicles	105000
10% Preference Capital	60000	Plant and machinery	235000
Equity shares Capital	300000	Sundry debtors	90000
Reserve fund	240000	Land and buildings	440000
Bank over draft	60000	Furniture and fixtures	140000
Provision for taxation	78000	Stock	60000
15% Debentures	420000	Short term investments	75000
Creditors	36000	Cash in hand	25000
Rent outstanding	6000	Cash at bank	30000
	1200000		1200000

EBIT = Rs. 204000

Solution:

$$\text{Debt-Equity Ratio} = \frac{\text{Debt}}{\text{Equity}} = \frac{\text{Outsider's funds}}{\text{Insider's funds}} = \frac{\text{Long term loans}}{\text{Share holders funds/Net worth}}$$

Debt = Debentures = 420000

Equity = Equity capital + Reserve fund

= 300000 + 240000 = 540000

$$\text{Debt-Equity Ratio} = \frac{420000}{540000} = 0.77:1$$

$$\text{Interest Coverage Ratio} = \frac{\text{EBIT}}{\text{Interest}} = \frac{\text{PBIT}}{\text{Interest}}$$

Interest = Debentures x 15/100 = 420000 x 15/100 = 63000

$$\text{Interest Coverage Ratio} = \frac{204000}{63000} = 3.23 \text{ times}$$

$$\text{Proprietors' funds to Total Assets Ratio} = \frac{\text{Proprietors' Funds}}{\text{Total Assets}}$$

Proprietors' Funds = Preference share capital + Equity share capital + Reserve fund

= 60000 + 300000 + 240000

= 600000

Total Assets = 1200000

$$\text{Proprietors' funds to Total Assets Ratio} = \frac{600000}{1200000} \times 100 = 50\%$$

Example 6: Calculate Gross Profit Margin, Net Operating Margin and Operating Ratio given the following information.

Sales	1000000	Cost of goods sold	600000
Selling and Administrative costs	200000	Depreciation	100000

Solution:

Gross Profit Margin = Sales – Cost of Goods Sold = 1000000 – 600000 = 400000

Net Profit Margin = Gross Profit – Selling and Administration costs – Depreciation
= 1000000 – 600000 = 400000

$$\text{Operating Ratio} = \frac{\text{Operating Cost}}{\text{Net Sales}}$$

Operating Cost = (Cost of goods sold + Operating expenses)
= (600000 + 200000 + 100000) = 900000

$$\text{Operating Ratio} = \frac{\text{Operating Cost}}{\text{Net Sales}} \times 100 = \frac{900000}{1000000} \times 100 = 90\%$$

Example 7: prepare a balance sheet from the following particulars.

Stock velocity	: 6
Gross profit margin	: 20%
Capital turnover ratio	: 2
Fixed assets turnover	: 4
Debt collection period	: 2 months
Creditors payment period	: 73 days
Gross profit	: Rs. 60000

Excess of closing stock over opening was : Rs. 5000

Solution

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Net Sales}}$$

$$20\% = \frac{60000}{\text{Net Sales}}$$

Net Sales = Gross Profit x Gross Profit Ratio = 60000 x 20% = 300000

$$\text{Stock Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Average Stock}}$$

Cost of Goods Sold = Net Sales – Gross Profit = 300000-60000=240000

$$\text{Average Stock} = \frac{\text{Cost of Goods Sold}}{\text{Stock Turnover Ratio}} = \frac{240000}{6} = 40000$$

$$\text{Average Stock} = \frac{\text{Opening Stock} + \text{Closing Stock}}{2}$$

Let us suppose that Opening Stock is 'x', then

$$40000 = \frac{x + (x + 5000)}{2}$$

$$40000 \times 2 = 2x + 5000$$

$$80000 - 5000 = 2x$$

$$x = \frac{75000}{2} = 37500$$

Opening Stock = 37500

Closing Stock = Opening Stock + 5000

Closing Stock = 37500 + 5000 = 42500

$$\text{Capital Turnover Ratio} = \frac{\text{Cost of sales}}{\text{Capital}}$$

$$\text{Capital} = \frac{\text{Cost of sales}}{\text{Capital Turnover Ratio}}$$

$$\text{Capital} = \frac{240000}{2} = 12000$$

$$\text{Fixed Assets Turnover Ratio} = \frac{\text{sales}}{\text{Fixed Assets}}$$

$$\text{Fixed Assets} = \frac{\text{sales}}{\text{Fixed Assets Turnover Ratio}}$$

$$\text{Capital} = \frac{300000}{4} = 75000$$

$$\text{Debtors Collection Period} = \frac{\text{365 Days/12 Months}}{\text{Debtors Turnover Ratio}}$$

$$\text{Debtors Turnover Ratio} = \frac{\text{12 Months}}{\text{Debt Collection Period}} = \frac{12}{2} = 6$$

$$\text{Debtors Turnover Ratio} = \frac{\text{Net Credit sales/Sales}}{\text{Average Debtors/Debtors}}$$

$$\text{Debtors} = \frac{\text{Sales}}{\text{Debtors Turnover Ratio}} = \frac{30000}{6} = 50000$$

$$\text{Creditors Payment Period} = \frac{\text{365 Days/12 Months}}{\text{Creditors Turnover Ratio}}$$

$$\text{Creditors Turnover Ratio} = \frac{\text{12 Months}}{\text{Creditors Payment Period}} = \frac{365}{73} = 5$$

$$\text{Creditors Turnover Ratio} = \frac{\text{Net Credit Purchases/Purchases}}{\text{Average Creditors/Creditors}}$$

Purchases = Sales +Closing Stock – Opening Stock – Gross Profit

Purchases = 300000 +42500 – 37500 – 60000 = 245000

$$\text{Creditors} = \frac{\text{Purchases}}{\text{Creditors Turnover Ratio}} = \frac{245000}{5} = 49000$$

Balance Sheet

Liabilities & Capital	Amount	Assets	Amount
Capital	120000	Fixed Assets	75000
Creditors	49000	Debtors	50000
		Closing Stock	42500
		Cash in hand(b/f)	1500
	169000		169000

USES OR ADVANTAGES OR IMPORTANCE OF RATIO ANALYSIS

Ratio Analysis stands for the process of determining and presenting the relationship of items and groups of items in the financial statements. It is an important technique of financial analysis. It is a way by which financial stability and health of a concern can be judged. The following are the main uses of Ratio analysis:

(a) Useful in financial position analysis: Accounting reveals the financial position of the concern. This helps banks, insurance companies and other financial institution in lending and making investment decisions.

(ii) Useful in simplifying accounting figures: Accounting ratios simplify, summaries and systematic the accounting figures in order to make them more understandable and in lucid form.

(iii) Useful in assessing the operational efficiency: Accounting ratios helps to have an idea of the working of a concern. The efficiency of the firm becomes evident when analysis is based on accounting ratio. This helps the management to assess financial requirements and the capabilities of various business units.

(iv) Useful in forecasting purposes: If accounting ratios are calculated for number of years, then a trend is established. This trend helps in setting up future plans and forecasting.

(v) Useful in locating the weak spots of the business: Accounting ratios are of great assistance in locating the weak spots in the business even through the overall performance may be efficient.

(vi) Useful in comparison of performance: Managers are usually interested to know which department performance is good and for that he compare one department with the another department of the same firm. Ratios also help him to make any change in the organisation structure.

LIMITATIONS OF RATIO ANALYSIS

1. **False results if based on incorrect accounting data**: Accounting ratios can be correct only if the data (on which they are based) is correct. Sometimes, the information given in the financial statements is affected by window dressing, i. e. showing position better than what actually is.
2. **No idea of probable happenings in future**: Ratios are an attempt to make an analysis of the past financial statements; so they are historical documents. Now-a-days keeping in view the complexities of the business, it is important to have an idea of the probable happenings in future.
3. **Variation in accounting methods**: The two firms' results are comparable with the help of accounting ratios only if they follow the some accounting methods or bases. Comparison will become difficult if the two concerns follow the different methods of providing depreciation or valuing stock.

4. **Price level change**: Change in price levels make comparison for various years difficult.
5. Only one method of analysis: Ratio analysis is only a beginning and gives just a fraction of information needed for decision-making so, to have a comprehensive analysis of financial statements, ratios should be used along with other methods of analysis.
6. **No common standards**: It is very difficult to by down a common standard for comparison because circumstances differ from concern to concern and the nature of each industry is different.
7. **Different meanings assigned to the some term:** Different firms, in order to calculate ratio may assign different meanings. This may affect the calculation of ratio in different firms and such ratio when used for comparison may lead to wrong conclusions.
8. **Ignores qualitative factors**: Accounting ratios are tools of quantitative analysis only. But sometimes qualitative factors may surmount the quantitative aspects. The calculations derived from the ratio analysis under such circumstances may get distorted.
9. **No use if ratios are worked out for insignificant and unrelated figure:** Accounting ratios should be calculated on the basis of cause and effect relationship. One should be clear as to what cause is and what effect is before calculating a ratio between two figures.

FUNDS FLOW ANALYSIS

INTRODUCTION

Balance sheet and profit and loss A/C show the financial status and profitability of the firm respectively. Balance sheet discloses the value of fixed assets as well as current assets, the decrease or increase of all assets and liabilities can be ascertained by comparing with the previous balance sheet. But it does not disclose the reasons for increasing or decreasing the assets/liabilities. However, the " Funds flow statement" is to be prepared to know such reasons. In this concept fund means " net working".

What is 'FUND' and 'FLOW' ?

A layman can describe word FUND as cash or cash equivalents. In technical terms, the word **FUND means** 'Net Working Capital'.

Funds may mean:

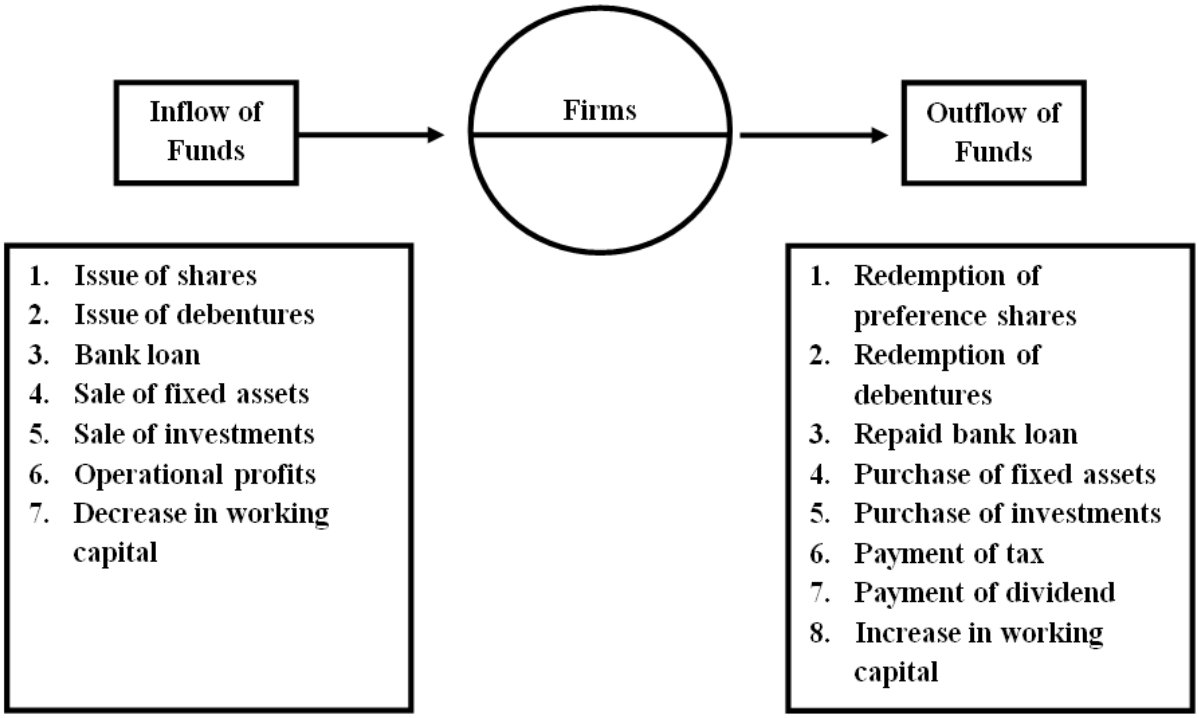
- Cash only
- Net working capital, i.e. current assets less current liabilities
- Total resources or total funds

- Internal resources only
- Net worth, i.e. owner's equity capital plus reserves

The term **FLOW refers to** change or transfer. The term 'Fund flow' or 'Flow of funds' may thus mean transport of:

- One asset to another
- One liability to another
- Assets to liabilities or vice versa
- The changes in working capital is also an inflow or outflow of funds, and thus it is called fund flow.

" Fund" is considered as working capital while preparing " Funds flow statement". Fund flow means change in the working capital. In other words, any increase or decrease in working capital means "flow of funds". Any transaction which has one current account and the other non-current account results in change in the working capital.



Current accounts:- Current assets accounts and current liabilities accounts are called current accounts. Assets which are converted into cash within a year are called current assets. Liabilities which are to be paid within a year are called current liabilities.

Non-current account:- Accounts which are not current accounts are called non-current accounts. For example, fixed assets accounts long term liabilities accounts, and capital & reserves accounts.

Preparation of funds flow statements

Funds flow statement is prepared by observing the items taken place during the periods of two balance sheets along with the adjustments into consideration.

Three statements are to be prepared.

I. Statement of changes in working capital
II. Statement of funds from operations
III. Funds flow statement

I. Statement of changes in working capital:- this statement reveals the net change in the working capital (CA – CL) . Current assets and current liabilities are as follows.

Current Assets	Current Liabilities
1.Cash in hand	1. Bills payables
2. Cash at bank	2. Creditors
3. Bills receivables	3. Outstanding expenses
4. Debtors	4. Dividends to be paid
5. Short term advances	5. Bank over draft
6. Short term investments	6. Provisions for current assets (ex: RBD)
7. Inventories or stock	7. Short-term loans
8. Incomes to be received	8. Taxes to be paid
9. Loose tools, etc.	9. Provision for tax (current liability or appropriation of profits)
10. Prepaid expenses	10. Proposed dividend (current liability or appropriation of profits), etc.

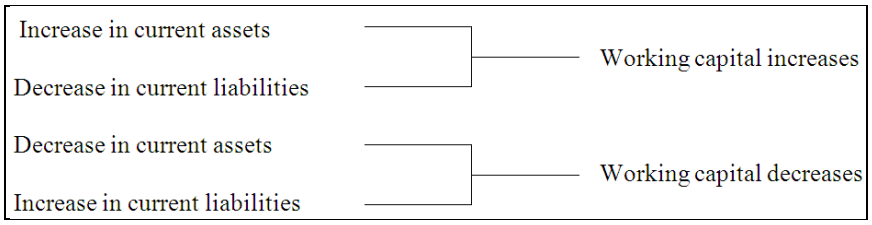
FORMAT OF STATEMENT OF CHANGES IN WORKING CAPITAL

Particulars	Previous year	Current year	Working capital	
			Increase	Decrease
A. Current Assets				
Cash in hand	xxx	xxx		
Cash at bank	xxx	xxx		
Bills receivables	xxx	xxx		
Debtors etc	xxx	xxx		
A	xxx	xxx		
B. Current Liabilities				
Bills payables	xxx	xxx		
Creditors	xxx	xxx		
Bank overdraft etc.	xxx	xxx		
B	xxx	xxx		

Working capital (A – B)	xxx	xxx		
Increase/Decrease in working capital	xxx	xxx		
B/F				
	xxx	xxx	xxx	xxx

Procedure for preparation

1. The figures of current assets and current liabilities of two balance sheets given are recorded in the respective columns provided. Don't take the adjustments. Then find working capital and increase/decrease in working capital.
2. By observing the current assets and current liabilities, the differences are shown in the working capital columns against to the each account and balance those columns. Increase or decrease in working capital can be recognized as follows:



II. Statement of funds from operations

FORMAT OF FUNDS FROM OPERATIONS

Particulars		Rs	Rs
Net profit			xxx
Add:			
	Provision for depreciation	xxx	
	Amortization of goodwill, patents, etc.	xxx	
	Preliminary expenses	xxx	
	Loss on sale of investments	xxx	
	Loss on sale of fixed assets	xxx	
	Provision for tax	xxx	
	Discount on issues of debentures	xxx	xxx
Less:			
	Dividend from investments	xxx	
	Profit on sale of investments	xxx	
	Profit on sale of fixed assets	xxx	
	Interest received	xxx	xxx
	Funds from operations		xxx

Procedure for preparation

1. Prepare the accounts for the items given in the adjustments to know the hidden information. Generally, the hidden information is as follows:

a) Purchase price of a fixed asset
b) Sale value of a fixed asset
c) Profit/loss on sale of a fixed asset
d) Depreciation provided on a fixed asset
e) Depreciation of the asset sold
f) Provision for tax for the year
g) Tax paid during the year
h) Dividend proposed during the year
i) Dividend paid during the year

2) By observing balance sheets and accounts for adjustments, non-Operating expenses and non-cash payment items (debit items of profit and loss account) are to be added to given net profit and all non-operating incomes & non-cash received items (credit items of profit and loss A/C) are to be deducted from given net profit to find the funds from operations.

Dr **Fixed Asset A/C** Cr

Particulars	Amount	Particulars	amount
To Balance b/d	***	By Depreciation	***
(opening)	***	(for asset sold)	
To P/L A/C (profit)	***	By Bank (sale value)	***
To Bank (purchase)		By P/L A/C (loss)	***
		By Balance c/d	***
	***		***

Dr **Cumulative Depreciation A/C** Cr

Particulars	Amount	Particulars	amount
To Fixed Asset A/C	***	By Balance b/d	***
(Depreciation of Asset		(opening)	***
sold)	***	By P/L A/C (provided)	
To Balance c/d (closing)	***		***

Dr **Provision for Tax A/C** Cr

Particulars	Amount	Particulars	amount
To Bank A/C (paid)	***	By Balance b/d	***
To Balance c/d (closing)	***	(opening)	***
	***	By P/L A/C (provided)	***

Note:- When tax paid is not given, it is considered that the tax provided in the previous year is paid in the current year.

Dr **Proposed Dividend A/C** Cr

Particulars	Amount	Particulars	amount
To Bank A/C (paid)	***	By Balance b/d	***
To Balance c/d (closing)	***	(opening)	***
	***	By P/L A/C (provided)	***

Note:- When dividend paid is not given, it is considered that the dividend proposed in the previous year is paid in the present year.

Note:- It can be found even if any one of the items is not given in the above said accounts.

2. In case provision for tax and proposed dividend are taken as current liabilities, there is no need to prepare those accounts. These should be shown in the " statement of changes in working capital" only.

   If these both are taken as an appropriation of profit (unlike current liabilities), there is a need to prepare their accounts to know the hidden information. Provision for tax and proposed dividend are to be added to the net profit to know the funds from operations and tax paid & dividend paid are shown on the applications' side in the funds flow statement.

III. Funds Flow Statement

FORMAT OF FUNDS FLOW STATEMENT

Particulars	Amount
SOURCES OF FUND	
Issue of shares	xxx
Issue of debentures	xxx
Sale of investments	xxx
Long-term loans	xxx
Non-operating incomes	xxx
Funds from operations	xxx
Decrease in working capital	xxx
Total Sources	xxx
APPLICATION OF FUND	
Redemption of pref. shares	xxx
Redemption of debentures	xxx
Purchase of investments	xxx
Purchase of fixed assets	xxx
Payment of long-term loans	xxx
Payment of tax	xxx
Payment of dividend	xxx
Increase in working capital	xxx
Total Uses	xxx

Procedure for preparation

1. By examining the non-current assets and liabilities, show as a source if cash comes and as an application if cash goes out.

2. Show non-operating income as a source and non-operating expenses as an application.
3. Through adjustment items, if cash comes in show as a source and if cash goes out show as an application.
4. If goodwill shows increase, take on the applications' side.

Note:- In order to say in single sentence, except the items taken in the " statements of changes in working capital and funds from operations", take as a source when cash comes through all other items and take as an application when cash goes out through all other items.

PRACTICE PROBLEMS

1. Prepare:
   1. Statement of changes in working capital and
   2. Funds flow statement from following balance sheets of Vijaya Mitra Ltd., as on 31-03-2016 and 31-03-2017:

Liabilities	**31-03-2016**	**31-03-2017**
Equity capital	200000	300000
Preference capital	200000	100000
Profit and loss	50000	75000
account	40000	60000
General resource	10000	50000
Unsecured loans	40000	5000
Current liabilities		
Total	540000	590000
Assets	**31-03-2016**	**31-03-2017**
Land and buildings	100000	80000
Plant and Machinery	90000	120000
Cash at Bank	60000	40000
Stock	120000	140000
Sundry Debtors	30000	50000
Vehicles	140000	160000
Total	540000	590000

Adjustments:
1. Dividend declared and paid Rs. 25000
2. Additional plant purchased Rs. 5000
3. Tax paid during the year Rs. 45000

Solution:

STATEMENT OF CHANGES IN WORKING CAPITAL

Particulars	Previous year	Current year	Working capital	
			Increase	Decrease
A. Current Assets				
Cash at bank	60000	40000		20000
Stock	120000	140000	20000	
Debtors	30000	50000	20000	
A	210000	230000		
B. Current Liabilities				
Current liabilities	40000	5000	35000	
B	40000	5000		
Working capital (A – B)	170000	225000		
Increase in working capital B/F	55000	-		55000
	225000	225000	75000	20000

FUNDS FROM OPERATIONS

Particulars	Amount	Amount
Net profit (75000-50000)		25000
Add:		
General reserve	20000	
Provision for dividend	25000	
Provision for tax	45000	
Depreciation on plant	20000	110000
Funds from operations		135000

FUNDS FLOW STATEMENT

Particulars	Amount
SOURCES OF FUND	
Issue of equity capital	100000
Unsecured loans procured	40000
Sale of land and building	20000
Funds from operations	135000
Total Sources	295000
APPLICATION OF FUND	
Redemption of pref. capital	100000
Purchase of vehicles	20000
Purchase of plant	50000
Dividend paid	25000
Tax paid	45000
Increase in working capital	55000
Total Uses	295000

Working notes:

Plant and Machinery A/C

Particulars	Amount	Particulars	Amount
To Opening balance	90000	By P&L A/C (Depn.) b/f	20000
To Bank (purchase)	50000	By Closing balance	120000
	140000		140000

Dividend A/C

Particulars	Amount	Particulars	Amount
To Bank A/C	25000	By P&L A/C (provi.) b/f	25000
	25000		25000

Tax A/C

Particulars	Amount	Particulars	Amount
To Bank A/C	45000	By P&L A/C (provi.) b/f	45000
	45000		45000

2. Prepare a funds flow statement from the following balance sheets

Liabilities	31-3-16	31-3-17	Assets	31-3-16	31-3-17
E.S.C	100000	200000	Machinery	120000	160000
P.S.C	170000	180000	Furniture	240000	140000
P & L A/C	260000	350000	Goodwill	12000	4000
G.R	110000	230000	Patents	8000	1000
9% Debentures	60000	140000	Cash	415000	1023000
Creditors	80000	200000	Preliminary	5000	2000
Bills payable	20000	30000	exp.		
	800000	1330000		800000	1330000

1. Depreciation on Machinery Rs. 15000 and on Furniture Rs. 12000
2. Dividend paid Rs. 18000

Solution:

STATEMENT OF CHANGES IN WORKING CAPITAL

Particulars	Previous year	Current year	Working capital	
			Increase	Decrease
A. Current Assets				
Cash	415000	1023000	608000	
A	415000	1023000		
B. Current Liabilities				
Credotors	80000	200000		120000
Bills payables	20000	30000		10000
B	100000	230000		
Working capital (A – B)	315000	793000		
Increase in working capital B/F	478000	-		478000
	793000	793000	608000	608000

FUNDS FROM OPERATIONS

Particulars	Amount	Amount
Net profit (350000-260000)		90000
Add:		
General reserve	120000	
Provision for dividend	18000	
Depreciation on furniture	12000	
Depreciation on machinery	15000	
Goodwill written off	8000	
Patents written off	7000	
Preliminary expenses written off	3000	183000
Funds from operations		273000

FUNDS FLOW STATEMENT

Particulars	Amount
SOURCES OF FUND	
Issue of equity capital	100000
Issue of preference capital	10000
Issue of debentures	80000
Sale of furniture	88000
Funds from operations	273000
Total Sources	551000
APPLICATION OF FUND	
Purchase of machinery	55000
Dividend paid	18000
Increase in working capital	478000
Total Uses	551000

Working Notes:

Machinery A/C

Particulars	Amount	Particulars	Amount
To Opening balance	120000	By P&L A/C (Depn.)	15000
To Bank (purchase)			
b/f	55000	By Closing balance	160000
	175000		175000

Furniture A/C

Particulars	Amount	Particulars	Amount
To Opening balance	240000	By P&L A/C (Depn.)	12000
		By Bank (Sale) b/f	88000
		By Closing balance	140000
	240000		240000

Dividend A/C

Particulars	Amount	Particulars	Amount
		By P&L A/C (provi.)	
To Bank A/C	18000	b/f	18000
	18000		18000

3. Balance sheets of M/s. Divya as on 1st January 2016 and 1st January 2017 were as follows:

Liabilities	2016	2017	Assets	2016	2017
Creditors	40000	44000	Cash	12000	27000
Overdraft	2000	3000	Debtors	30000	50000
Long term loan	40000	50000	Stock	35000	25000
Capital	125000	150000	Machinery	80000	55000
Reserves	10000	10000	Land	40000	50000
P& L	15000	30000	building	35000	80000
	232000	287000		232000	287000

During the year machine costing Rs. 10000 (accumulated depreciation Rs. 3000) was sold for Rs. 5000.

Solution:

STATEMENT OF CHANGES IN WORKING CAPITAL

Particulars	Previous year	Current year	Working capital	
			Increase	Decrease
A. Current Assets				
Cash	12000	27000	15000	
Debtors	30000	50000	20000	
Stock	35000	25000		10000
A	77000	102000		
B. Current Liabilities				
Credotors	40000	44000		4000
Overdraft	2000	3000		1000
B	42000	47000		
Working capital (A – B)	35000	55000		
Increase in working capital B/F	20000			20000
	55000	55000	35000	35000

FUNDS FROM OPERATIONS

Particulars	Rs	Rs
Net profit (30000-15000)		15000
Add:		
General reserve		
Provision for dividend		
Depreciation on furniture		
Depreciation on machinery	18000	
Loss on sale of machinery	2000	20000
Funds from operations		35000

FUNDS FLOW STATEMENT

Particulars	Amount
SOURCES OF FUND	
Issue of capital	25000
Long-term loan	10000
Sale of machinery	5000
Funds from operations	35000
Total Sources	75000
APPLICATION OF FUND	
Purchase of land	10000
Purchase of building	45000
Increase in working capital	20000
Total Uses	75000

Working Notes:

Machinery A/C

Particulars	Amount	Particulars	Amount
To Opening balance	80000	By P&L A/C (Depn.)	3000
		By P&L A/C (loss on sale)	2000
		By Bank (sale)	5000
		By Closing balance	55000
		By P&L A/C (Depn.) b/f	15000
	80000		80000

4. Prepare a funds flow statement from the following:

Liabilities	2002	2003	Assets	2002	2003
Share capital	200000	250000	Cash	30000	47000
Creditors	70000	45000	Debtors	120000	115000
Retained earnings	10000	23000	Stock	80000	90000
			Land	50000	66000
	280000	318000		280000	318000

Solution:

STATEMENT OF CHANGES IN WORKING CAPITAL

Particulars	Previous year	Current year	Working capital	
			Increase	Decrease
A. Current Assets				
Cash	30000	47000	17000	
Debtors	120000	115000		5000
Stock	80000	90000	10000	
A	230000	252000		
B. Current Liabilities				
Creditors	70000	45000	25000	
B	70000	45000		
Working capital (A – B)	160000	207000		
Increase in working capital B/F	47000	-		47000
	207000	207000	52000	52000

FUNDS FROM OPERATIONS

Particulars	Rs	Rs
Net profit (23000-10000)		13000
Funds from operations		13000

FUNDS FLOW STATEMENT

Particulars	Amount
SOURCES OF FUND	
Issue of capital	50000
Funds from operations	13000
Total Sources	63000
APPLICATION OF FUND	
Purchase of land	16000
Increase in working capital	47000
Total Uses	63000

5. From the information given below, prepare funds flow statement of Global Co. Ltd.

Liabilities	I year	II year	Assets	I year	II year
Share capital			Goodwill	190000	140000
Equity capital	450000	600000	Plant	160000	250000
Preference capital	225000	150000	Building	240000	195000
Profit and loss A/C	60000	75000	Inventories	92000	235000
Proposed dividend	55000	67000	trade	175000	125000
Trade creditors	72000	90000	Debtors	45000	57000
Bills payable	32000	25000	Bills receivables	52000	77000
Provision for tax	60000	72000	cash		
	954000	1079000		954000	1079000

Additional information:

1. An interim dividend of Rs. 35000 has been paid in II year.
2. Payment of income tax Rs. 52000 was paid during II year.
3. Depreciation of Rs. 35000 and Rs. 42000 have been charged on plant and building respectively in II year.
4. A part of the plant worth Rs. 20000was sold for Rs. 30000.

Solution:

STATEMENT OF CHANGES IN WORKING CAPITAL

Particulars	Previous year	Current year	Working capital	
			Increase	Decrease
A. Current Assets				
Cash	52000	77000	25000	
Inventories	92000	235000	143000	
Debtors	175000	125000		50000
Bills receivables	45000	57000	12000	
A	364000	494000		
B. Current Liabilities				
Credotors	72000	90000		18000
Bills payables	32000	25000	7000	
B	104000	115000		
Working capital (A – B)	260000	379000		
Increase in working capital B/F	119000			119000
	379000	379000	187000	187000

FUNDS FROM OPERATIONS

Particulars	Rs	Rs
Net profit (75000-60000)		15000
Add:		
Goodwill written off	50000	
Provision for dividend	47000	
Provision for tax	64000	
Depreciation on plant	35000	
Depreciation on building	42000	238000
Less:		253000
Profit on sale of plant	10000	10000
Funds from operations		243000

FUNDS FLOW STATEMENT

Particulars	Amount
SOURCES OF FUND	
Issue of equity capital	150000
Sale of building	3000
Sale of plant	30000
Funds from operations	243000
Total Sources	426000
APPLICATION OF FUND	
Redemption of preference shares	75000
Purchase of plant	145000
Dividend paid	35000
Tax paid	52000
Increase in working capital	119000
Total Uses	426000

Working Notes:

Plant A/C

Particulars	Amount	Particulars	Amount
To Opening balance	160000	By P&L A/C (Depn.)	35000
To P & L A/C (porofit)	10000	By Bank (Sale)	30000
To Bank (purchase) b/f	145000	By Closing balance	250000
	315000		315000

Building A/C

Particulars	Amount	Particulars	Amount
To Opening balance	240000	By P&L A/C (Depn.)	42000
		By Bank (Sale) b/f	3000
		By Closing balance	195000
	240000		240000

Proposed dividend A/C

Particulars	Amount	Particulars	Amount
To Bank A/C	35000	By Opening balance	55000
To Closing balance	67000	By P&L A/C (provi.) b/f	47000
	102000		102000

Provision for tax A/C

Particulars	Amount	Particulars	Amount
To Bank A/C	52000	By Opening balance	60000
To Closing balance	72000	By P&L A/C (provi.) b/f	64000
	124000		124000

ADVANTAGES OF FUND FLOW STATEMENTS

1. Funds flow statement reveals the net result of Business operations done by the company during the year.

2. In addition to the balance sheet, it serves as an additional reference for many interested parties like analysts, creditors, suppliers, government to look into financial position of the company.

3. The Fund Flow Statement shows how the funds were raised from various sources and also how those funds were deployed by a company.

4. It reveals the causes for the changes in liabilities and assets between the two balance sheet dates.

5. Funds flow statement helps the management in deciding its future course of plans and also it acts as a control tool for the management.

Disadvantages of Fund Flow Statements

1. Funds Flow statement has to be used along with balance sheet and profit and loss account for inference of financial strengths and weakness of a company it cannot be used alone.

2. Fund Flow Statement does not reveal the cash position of the company, and that is why company has to prepare cash flow statement in addition to funds flow statement.

3. Funds flow statement only rearranges the data which is there in the books of account and therefore it lacks originality.

4. Funds flow statement is basically historic in nature, that is it indicates what happened in the past and it does not communicate anything about the future, only estimates can be made based on the past data and therefore it cannot be used the management for taking decision related to future.

BEFA UNIT V
CASH FLOW ANALYSIS

INTRODUCTION

A **Cash Flow Statement** is a statement showing changes in cash position of the firm from one period to another. It explains the inflows (receipts) and outflows (disbursements) of cash over a period of time. The inflows of cash may occur from sale of goods, sale of assets, receipts from debtors, interest, dividend, rent, issue of new shares and debentures, raising of loans, short-term borrowing, etc. The cash outflows may occur on account of purchase of goods, purchase of assets, payment of loans loss on operations, payment of tax and dividend, etc

The cash flow statement shows how much cash comes in and goes out of the company over the quarter or the year. At first glance, that sounds a lot like the income statement in that it records financial performance over a specified period. But there is a big difference between the two.

What distinguishes the two is accrual accounting, which is found on the income statement. Accrual accounting requires companies to record revenues and expenses when transactions occur, not when cash is exchanged. At the same time, the income statement, on the other hand, often includes non-cash revenues or expenses, which the statement of cash flows does not include.

Just because the income statement shows net income of Rs.10 does not mean that cash on the balance sheet will increase by Rs.10. Whereas when the bottom of the cash flow statement reads Rs.10 net cash inflow, that's exactly what it means. The company has Rs.10 more in cash than at the end of the last financial period. You may want to think of net cash from operations as the company's "true" cash profit.

Because it shows how much actual cash a company has generated, the statement of cash flows is critical to understanding a company's fundamentals. It shows how the company is able to pay for its operations and future growth.

Indeed, one of the most important features you should look for in a potential investment is the company's ability to produce cash. Just because a company shows a profit on the income statement doesn't mean it cannot get into trouble later because of insufficient cash flows. A close examination of the cash flow statement can give investors a better sense of how the company will fare.

OBJECTIVES OF CASH FLOW STATEMENT

1. Highlighting cash flow from different activities
2. Short-term Planning
3. Cash Flow information helps to understand liquidity
4. Efficient cash management
5. Prediction of sickness
6. Comparison with budget
7. Cash position

CURRENT ASSESTS AND CURRENT LIABILITIES

CURRENT ASSETS	CURRENT LIABILITIES
1. Cash in hand	1. Bills payables
2. Cash at bank	2. Creditors
3. Bills receivables	3. Outstanding expenses
4. Debtors	4. Dividends to be paid

5. Short term advances	5. Bank over draft
6. Short term investments	6. Short term loans
7. Inventories or stock	7. Provisions for current assets (ex: RBD)
8. Pre- paid expenses	8. Taxes to be paid
9. Incomes to be received	9. Provision for tax
10. Loose tools, etc.	10. Proposed dividend

The general rules that develop from the above discussion are:

1. An **increase** in current assets leads to **decrease** in cash.

2. A **decrease** in current assets leads to an **increase** in cash.

3. An **increase** in current liabilities leads to an **increase** in cash.

4. A **decrease** in current liabilities leads to a **decrease** in cash.

PREPARING A CASH FLOW STATEMENT

Cash flow statements have three distinct sections, each of which relates to a particular component – operations, investing and financing – of a company's business activities.

The indirect method of presentation is very popular, because the information required for it is relatively easily assembled from the accounts that a business normally maintains in its chart of accounts. The indirect method is less favoured by the standard-setting bodies, since it does not give a clear view of how cash flows through a business. The alternative reporting method is the direct method.

Three Sections of the Cash Flow Statement

Companies produce and consume cash in different ways, so the cash flow statement is divided into three sections: cash flows from operations, financing and investing. Basically, the sections on operations and financing show how the company gets its cash, while the investing section shows how the company spends its cash.

1. **Operating Activities:** Operating activities include cash flows from all standard business operations. Cash receipts from selling goods and services represent the inflows. The revenues from interest and dividends are also included here. The operational expenditures are considered as outflows for this section. Although interest expenses fall under this section but the dividends are not included .Dividends are considered as a part of financing activity in financial accounting terms.
2. **Investing Activities:** Investing activities include transactions with assets, marketable securities and credit instruments. The sale of property, plant and equipment or marketable securities is a cash inflow. Purchasing property, plant and equipment or marketable securities are considered as cash outflows. Loans made to borrowers for long-term use is another cash outflow. Collections from these loans, however, are cash inflows.

3. **Financing Activities:** Financing activities on the statement of cash flows are much more defined in nature. The receipts come from borrowing money or issuing stock. The outflows occur when a company repays loans, purchases treasury stock or pays dividends to stockholders. As the case with

other activities on the statement of cash flows depend on activities rather than actual general ledger accounts.

CASH FLOW STATEMENT FORMAT

Cash Flow Statement of ---------- Company for the year ended -------

Particulars	Rs.	Rs.
A. Cash Flow from Operating Activities		
Net Profit	xxx	
Adjustments: (to convert net profit to cash provided by operating activities)		
Goodwill written off	xxx	
Loss on sale of investments	xxx	
Loss on sale of fixed assets	xxx	
Preliminary expenses written off	xxx	
Discount on issue of debentures	xxx	
Provision for tax	xxx	
Patents written off	xxx	
Transfer to reserve	xxx	
Provision for dividend	xxx	
Profit on sale of fixed assets/investments	(xxx)	
Profit from investments	(xxx)	
Increase in current liabilities	xxx	
Decrease in current liabilities	(xxx)	
Increase in current assets	(xxx)	
Decrease in current assets	xxx	
Cash from operations		xxx
B. Cash Flows from Investing Activities		
Purchase of fixed assets	(xxx)	
Proceeds from sale of fixed assets	xxx	
Purchase of long-term investments	(xxx)	
Proceeds from sale of long-term investments	xxx	
Cash from investing activities		xxx
C. Cash Flows from Financing Activities		
Issue of shares	xxx	
Issue of debentures	xxx	
Share capital repaid	(xxx)	
Debentures repaid	(xxx)	
Cash from financing activities		xxx
Net increase/decrease in cash		xxx
Cash at the beginning of the year		xxx
Cash at the end of the year		xxx

Steps to know cash flow from Operating Activities

1) Prepare accounts for adjustments to find hidden information.
2) Show current year net profit (Closing balance – Opening balance)
3) Add Non – cash and Non – operating expenses by observing balance sheets and accounts for adjustments.
4) Deduct Non – cash and Non – operating incomes by observing balance sheets and accounts for adjustments.
5) Adjust current assets & current liabilities except cash and bank balances.

Steps to know cash flow from Investing Activities

1) Observe all fixed assets and their adjustment accounts.
2) Purchase of fixed assets and investments result in outflow of cash.
3) Sale of fixed assets and investments result in inflow of cash.

Steps to know cash flow from Financing Activities

1) Observe all capital & long-term liabilities and their adjustment accounts.
2) Increase in capital and long-term loans result in inflow of cash.
3) Decrease in capital and long-term loans result in outflow of cash.

ADVANTAGES OF CASH FLOW STATEMENT

1. It shows the actual cash position available with the company between the two balance sheet dates which funds flow and profit and loss account are unable to show. So it is important to make a cash flow report if one wants to know about the liquidity position of the company.

2. It helps the company in accurately projecting the future liquidity position of the company enabling it arrange for any shortfall in money by arranging finance in advance and if there is excess than it can help the company in earning extra return by deploying excess funds.

3. It acts like a filter and is used by many analyst and investors to judge whether company has prepared the financial statements properly or not because if there is any discrepancy in the cash position as shown by balance sheet and the cash flow statement, it means that statements are incorrect.

DISADVANTAGES OF CASH FLOW STATEMENT

1. Since it shows only cash position, it is not possible to deduce actual profit and loss of the company by just looking at this statement.

2. In isolation this is of no use and it requires other financial statements like balance sheet, profit and loss etc…, and therefore limiting its use.

PRACTICE PROBLEMS AND SOLUTIONS

1. From the following balance sheets as on 31-12-2016 and 31-12-2017. You are required to prepare a cash flow statement.

Liabilities	**31-12-2016**	**31-12-2017**
Share capital	100000	150000
Profit and loss A/C	50000	80000
General reserve	30000	40000
12% Bonds	50000	60000
Sundry creditors	30000	40000
Outstanding expenses	10000	15000
Total	270000	385000
Assets	**31-12-2016**	**31-12-2017**
Fixed assets	100000	150000
Goodwill	50000	40000
Inventories	50000	80000
Bank	10000	15000
Bills receivables	10000	20000
Sundry debtors	50000	80000
Total	270000	385000

Solution:

Cash Flow Statement for the year ended 31-12-2017

Particulars	**Rs.**	**Rs.**
A. Cash Flow from Operating Activities		
Net Profit (80,000 - 50,000)	30000	
Adjustments:		
Goodwill written off	10000	
Transfer to reserve	10000	
Increase in creditors	10000	
Increase in outstanding expenses	5000	
Increase in inventory	(30,000)	
Increase in receivables	(10,000)	
Increase in debtors	(30,000)	
Cash from operations		(5,000)
B. Cash Flows from Investing Activities		
Purchase of fixed assets	(50,000)	
Cash from investing activities		(50,000)
C. Cash Flows from Financing Activities		
Issue of shares	50,000	
Issue of bonds	10,000	
Cash from financing activities		60,000
Net increase in cash		5,000
Bank at the beginning of the year		10,000
Bank at the end of the year		15,000

2. Prepare a cash flow statement from the following balance sheets of Kumar Ltd.

Liabilities	2016	2017	Assets	2016	2017
Capital	150000	175000	Land & Building	110000	150000
Loan from bank	160000	100000	Machinery	200000	140000
Creditors	85000	93000	Stock	50000	45000
Outstanding expenses	5000	7000	Debtors	70000	80000
Bills payable	50000	40000	Cash	15000	22000
Long term loan	--------	25000	Pre – paid expenses	5000	3000
	450000	440000		450000	440000

Net profit during the year 2017 was Rs/ 60000. During 2017 machinery costing Rs. 25000 (accumulated depreciation Rs. 10000) was sold for Rs. 25000. The tax payable and dividend payable were Rs. 50000 and Rs. 35000 respectively during 2017.

Solution:

Cash Flow Statement for the year ended 31-12-2017

Particulars	Rs.	Rs.
A. Cash Flow from Operating Activities		
Net Profit	60000	
Adjustments:		
Depreciation on machinery	45000	
Profit on sale of machinery	(10000)	
Provision for tax	50000	
Provision for dividend	35000	
Increase in creditors	8000	
Increase in outstanding expenses	2000	
Decrease in bills payable	(10000)	
Decrease in stock	5000	
Increase in debtors	(10000)	
Decrease in prepaid expenses	2000	
Cash from operations		177000
B. Cash Flows from Investing Activities		
Purchase of land and building	(40000)	
Sale of machinery	25000	
Cash from investing activities		(15000)
C. Cash Flows from Financing Activities		
Bank loan repaid	(60000)	
Long-term loan repaid	25000	
Tax paid	(50000)	
Dividend paid	(35000)	
Drawings	(35000)	
Cash from financing activities		(155000)
Net increase/decrease in cash		7000
Cash at the beginning of the year		15000
Cash at the end of the year		22000

Working notes:

Note: For working notes given in the problem, accounts are to be prepared to extract the hidden information.

Machinery Account

Particulars	Amount	Particulars	Amount
To Opening balance	200000	By Bank (Sales)	25000
To P & L A/C (Profit)	10000	By Depreciation B/F	45000
		By Closing balance	140000
	210000		210000

Tax Account

Particulars	Amount	Particulars	Amount
To Bank Account	50000	By P & L A/C (provided) B/F	50000
	50000		50000

Dividend Account

Particulars	Amount	Particulars	Amount
To Bank Account	35000	By P & L A/C (provided) B/F	35000
	35000		35000

3. From the following balance sheets of 1993 and 1994, prepare the cash flow statement.

Liabilities	2016	2017	Assets	2016	2017
Equity capital	20000	25000	Plant	46000	45000
Debentures	15000	12000	Debtors	9000	7000
Creditors	16000	18000	Stock	5000	9000
Profit and loss A/C	11000	14000	Cash	2000	8000
	62000	69000		62000	69000

Solution:

Cash Flow Statement for the year ended 31-12-2017

Particulars	Rs.	Rs.
A. Cash Flow from Operating Activities		
Net Profit (14000-11000)	3000	
Adjustments:		
Increase in creditors	2000	
Decrease in debtors	2000	
Increase in stock	(4000)	
Cash from operations		3000
B. Cash Flows from Investing Activities		
Sale of plant	1000	
Cash from investing activities		1000
C. Cash Flows from Financing Activities		
Issue of equity capital	5000	
Debentures repaid	(3000)	

Cash from financing activities		2000
Net increase/decrease in cash		6000
Cash at the beginning of the year		2000
Cash at the end of the year		8000

4. Prepare a cash flow statement from the following balance sheets as on 31-12-2016 and 31-12-2017.

Liabilities	2016	2017	Assets	2016	2017
Share capital	100000	150000	Fixed assets	100000	150000
Profit and loss A/C	50000	80000	Goodwill	50000	40000
General reserve	30000	40000	Inventories	50000	80000
6% bonds	50000	60000	Debtors	50000	80000
Creditors	30000	40000	Bills receivables	10000	20000
Outstanding expenses	10000	15000	Bank	10000	15000
	270000	385000		270000	385000

Solution:

Cash Flow Statement for the year ended 31-12-2017

Particulars	Rs.	Rs.
A. Cash Flow from Operating Activities		
Net Profit (80000-50000)	30000	
Adjustments:		
General Reserve	10000	
Goodwill written off	10000	
Interest on bonds (50000x6/100)	3000	
Increase in creditors	10000	
Increase in outstanding expenses	5000	
Increase in inventory	(30000)	
Increase in debtors	(30000)	
Icrease in bills receivables	(10000)	
Cash from operations		(2000)
B. Cash Flows from Investing Activities		
Purchase of fixed assets	(50000)	
Cash from investing activities		(50000)
C. Cash Flows from Financing Activities		
Issue of equity capital	50000	
Issue of bonds	10000	
Interest on bonds paid	(3000)	
Cash from financing activities		57000
Net decrease in cash		5000
Bank at the beginning of the year		10000
Bank at the end of the year		15000

5. Financial statements of Mr. Ram are as follows

Liabilities	2016	2017	Assets	2016	2017
Bills payable	29000	25000	Cash	40000	30000
Capital	731000	615000	Debtors	20000	17000
			Stock	8000	13000
			Buildings	92000	80000
			machinery	600000	500000
	760000	640000		760000	640000

Additional information:

a) Proprietor has not taken any drawings.
b) There are no purchases and sales in buildings and machinery.

Prepare cash flow statement.

Cash Flow Statement for the year ended 31-12-2017

Particulars	Rs.	Rs.
A. Cash Flow from Operating Activities		
Net Profit	(116000)	
Adjustments:		
Depreciation on buildings	12000	
Depreciation on machinery	100000	
Decrease in bills payable	(4000)	
Decrease in debtors	3000	
Increase in stock	(5000)	
Cash from operations		(10000)
B. Cash Flows from Investing Activities		
	0	
Cash from investing activities		0
C. Cash Flows from Financing Activities		
	0	
Cash from financing activities		0
Net decrease in cash		(10000)
Cash at the beginning of the year		40000
Cash at the end of the year		30000

Working Notes:

Capital Account

Particulars	Amount	Particulars	Amount
To P & L A/C (loss) B/F	116000	By Opening balance	731000
To Closing balance	615000		
	731000		731000

DIIFERENCES BETWEEN FUNDS FLOW STATEMENT AND CASH FLOW STATEMENT

Table of Difference between Funds Flow Statement and Cash Flow Statement

	Basis of Difference	Cash Flow Statement	Funds Flow Statement
1.	**Meaning of fund**	Funds means only cash which is a component of net current assets.	Fund means net working capital (i.e. current assets minus current liabilities).
2.	**Objective**	Its objective is to know about the changes occurred in cash position between two balance sheet dates.	Its objective is to know about the changes occurred in net working capital between two balance dates.
3.	**Basis of preparation**	Increase in current liability or decrease in current asset (except cash) results in an increase in cash.	Increase in current liability and decrease in current asset results in a decrease in net working
4.	**Effect of transaction**	Effect of a transaction on cash is considered.	Effect of a transaction on net working capital is considered.
5.	**Utility**	Cash flow statement is useful for short-term analysis.	Fund flow statement is useful for long-term analysis
6.	**Statement of changes in Working Capital**	No such statement is prepared separately in cash flow statement.	A separate statement for changes in working capital is prepared in fund flow statement or analysis.
7	**Cash Balances**	Opening and closing balances of cash are shown in this statement	Such balances of cash are shown in statement of changes in working capital.